USC
Viterbi
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ISE529 Predictive Analytics
Instructor: Dr. Tao Ma
2025 Fall

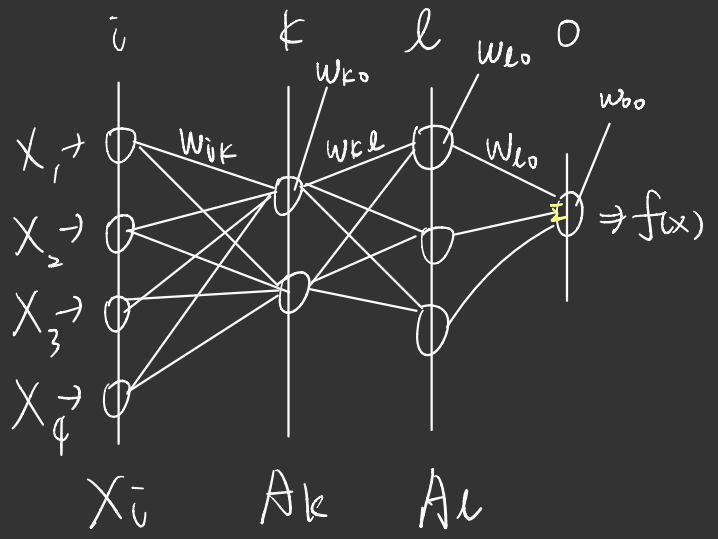
Review Session for Final Exam

1. Consider a neural network with two hidden layers: $p = 4$ input units, 2 units in the first hidden layer, 3 units in the second hidden layer, and a single output.
(a) Draw a picture of the network.
(b) Write out an expression for $f(X)$, assuming ReLU activation functions.
(c) How many parameters are there?

2. **Neural Network** calculation example on slides.

3. Suppose we produce ten bootstrapped samples from a data set containing red and green classes. We then apply a **classification tree** to each bootstrapped sample and, for a specific value of X, produce 10 estimates of $Pr(\text{Class is Red}|X)$:

(a)



(b)

$$g(z)\begin{cases} 0 & \text{if } z<0 \\ z & \text{if } z>0 \end{cases}$$

$$f(x) = g\left(w_{00} + \sum_{l=1}^{3} w_{l0} A_l\right)$$

$$A_l = g\left(w_{l0} + \sum_{k=1}^{2} w_{kl} A_k\right)$$

$$A_k = g\left(w_{k0} + \sum_{i=1}^{4} w_{ik} x_i\right)$$

ch4-22.

(c)

$4\times2 + 2\times3 + 2 + 3\times1 + 3 + 1 \Rightarrow 3.$

i的为 input　k的为 output　l的为 out

K. l. 0都有 intercept.

3. Suppose we produce ten bootstrapped samples from a data set containing red and green classes. We then apply a **classification tree** to each bootstrapped sample and, for a specific value of X, produce 10 estimates of $Pr(\text{Class is Red}|X)$:

0.2, 0.30, 0.35, 0.4, 0.15, 0.25, 0.75, 0.86, 0.9, and 0.85.

There are two common ways to combine these results together into a single class prediction. One is the majority vote approach. The second approach is to classify based on the average probability. What is the final classification under each of these two approaches?

4. **Decision tree example** on slides.

(3) ① Majority vote approach.

0.2 → g
0.3 → g
0.35 → g
0.4 → g
⋮

g = 6
r = 4.

75% →

② classify (average prob)

$\frac{\text{Sum}}{10} = 0.501$ X → Red.

choi - il 負.

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5. Suppose we estimate the regression coefficients in a linear regression model by minimizing

Lasso → $\sum_{i=1}^{n}\left(y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right)^2$ subject to $\sum_{j=1}^{p}|\beta_j|\le s$ $|\beta_0|+|\beta_1|\le t$.

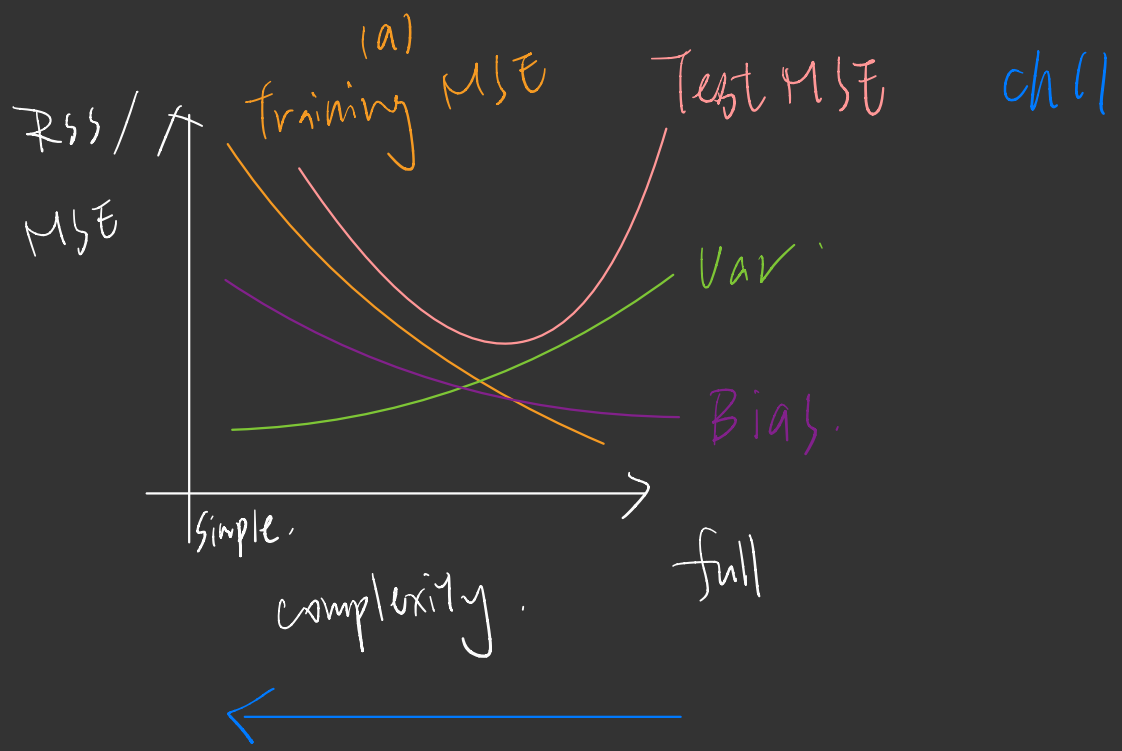
for a particular value of s. For parts (a) through (e), indicate which of i. through v. is correct. Justify your answer.

(a) As we increase s from 0, the training RSS will:

i. Increase initially, and then eventually start decreasing in an inverted U shape.
ii. Decrease initially, and then eventually start increasing in a U shape.
iii. Steadily increase.
iv. Steadily decrease.
v. Remain constant.

(b) Repeat (a) for test RSS.
(c) Repeat (a) for variance.
(d) Repeat (a) for (squared) bias.
(e) Repeat (a) for the irreducible error.

t increase → model simple.



6. Suppose we estimate the regression coefficients in a linear regression model by minimizing

Ridge → $\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2$

for a particular value of λ. For parts (a) through (e), indicate which of i. through v. is correct. Justify your answer.

(a) As we increase λ from 0, the training RSS will:
- i. Increase initially, and then eventually start decreasing in an inverted U shape.
- ii. Decrease initially, and then eventually start increasing in a U shape.
- iii. Steadily increase.
- iv. Steadily decrease.
- v. Remain constant.

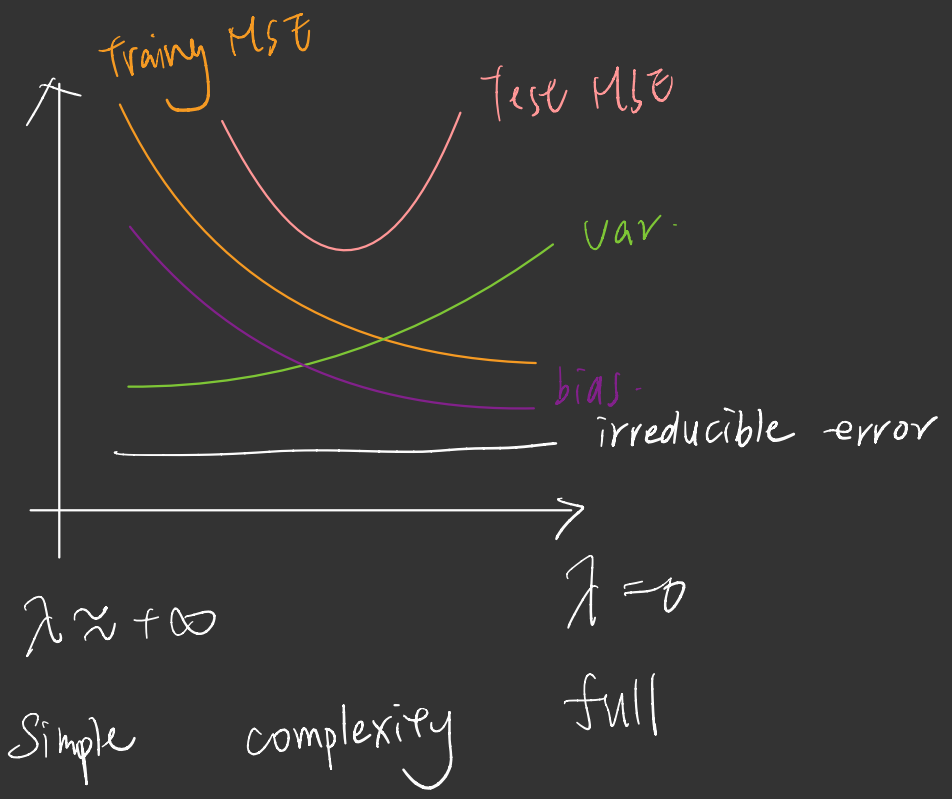
(b) Repeat (a) for test RSS.
(c) Repeat (a) for variance.
(d) Repeat (a) for (squared) bias.
(e) Repeat (a) for the irreducible error.

7. What is the essential difference between Lasso and Ridge methods? Why?

RSS
or
MSE

Sample
Null

(6)



ch11 p.45

(7)

① { Ridge : coefficient will just approach 0.
Lasso : coefficient could equal 0 aka parameter could be eliminated.

② constraint region { $|\beta_0| + |\beta_1| \le t$
$\beta_0^2 + \beta_1^2 \le t$.

The lasso constraint has corners at each of the axes, and so the ellipse will often intersect the constraint region at an axis. When this occurs, one of the coefficients will equal zero.